

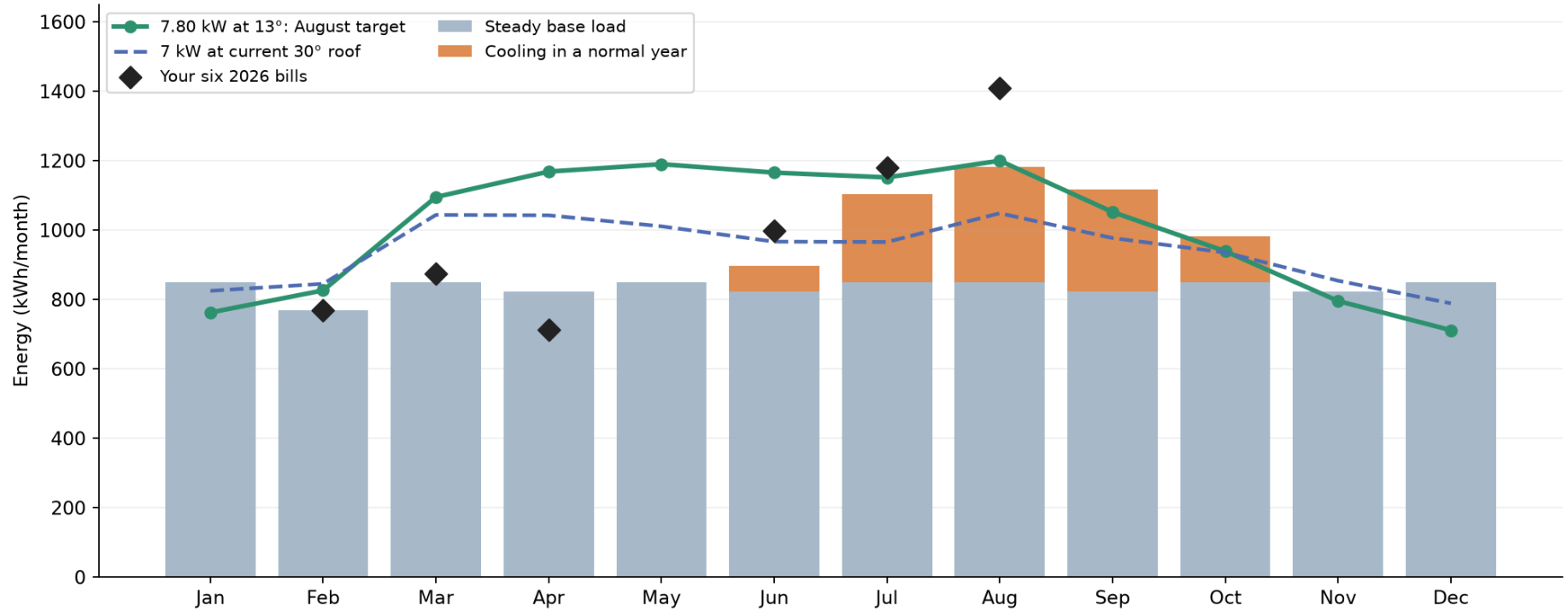
Garage Solar Sizing and Cost Optimization

ZIP 92109 • August target: 1,200 kWh • shade-free south-facing roof

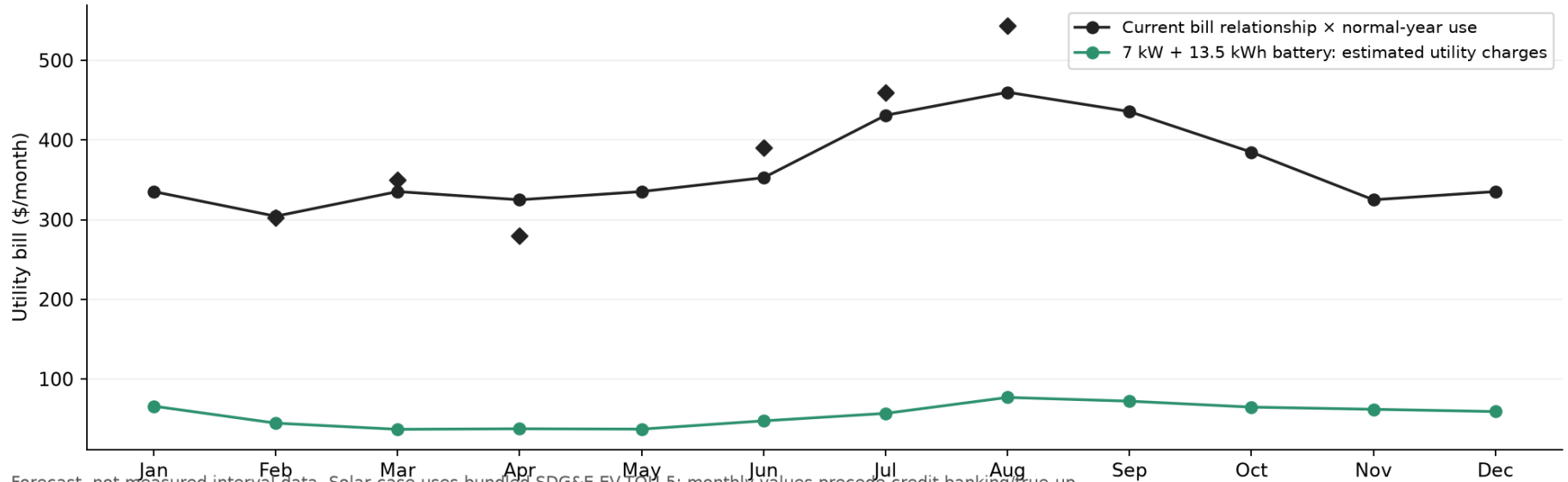
The practical answer is to keep the 30° roof. A 13–14° south-facing array is the mathematical August optimum per installed kilowatt, but it produces only 2.7% more in August and about 4.2% less over the full year than 30°. More seriously, the shallower roof loses sloped surface area over this fixed footprint. The present 30° roof needs about 8.01 kW to make 1,200 kWh in a typical August. That requires 391.9 sq ft of active 22%-efficient cells on the roughly 401.3 sq ft exposed solar slope: 97.7% active coverage. At 23% efficiency it requires 93.5% coverage. Therefore the target is possible only with a very tightly packed custom solar skin, higher-efficiency cells, a modest roof enlargement, or a steeper slope.

Economic result: under the central custom-roof cost assumptions, the model selects 6 kW plus one 13.5 kWh battery. Its first-year utility saving is about \$3,465, but after annualizing installation, maintenance and replacement costs over 25 years at a 5% real discount rate, the margin is only \$58/year. A target-sized 8 kW array with one battery barely lowers the bill beyond 7 kW because the extra midday energy is exported cheaply. If the custom roofing work belongs to the building budget anyway, the same 6 kW + battery case improves to about \$590/year of levelized net savings.

Your base load dominates the year; cooling drives the summer increase



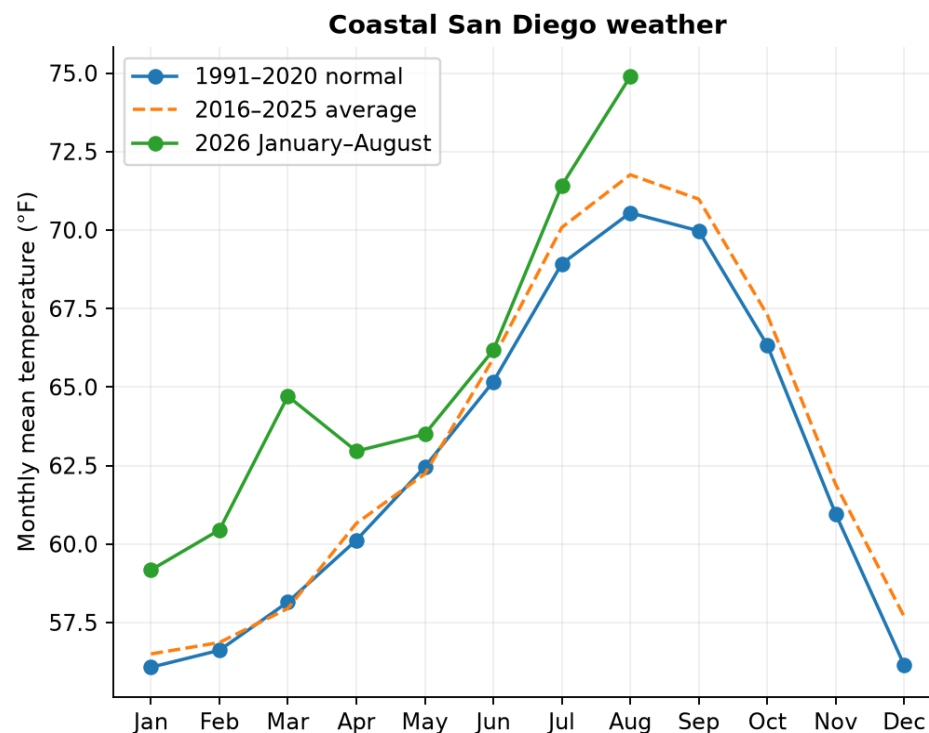
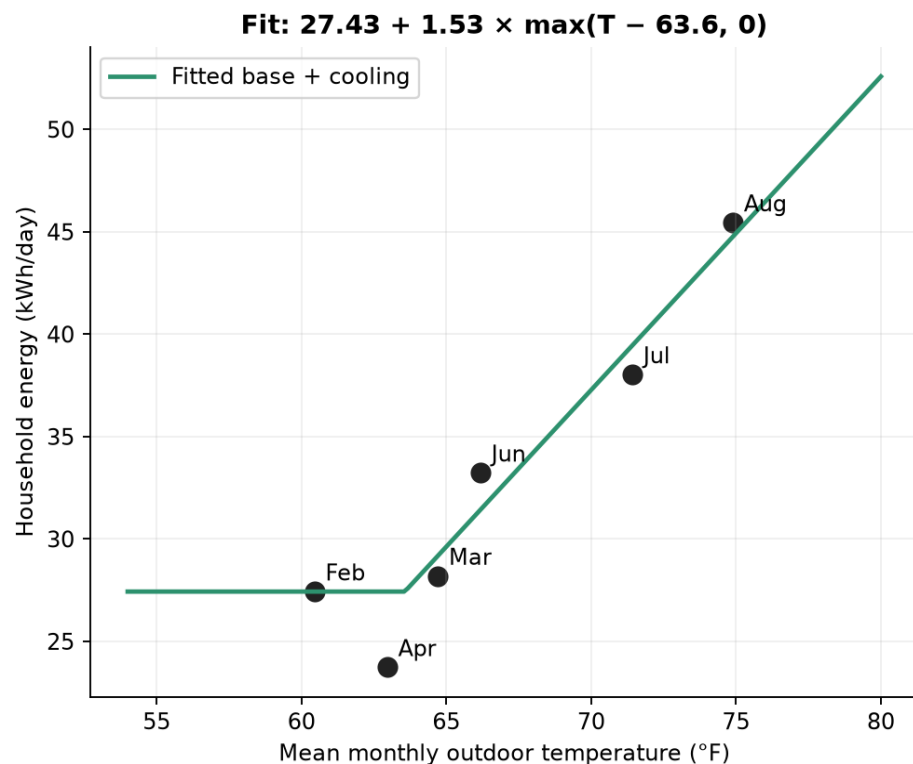
Utility charges only — equipment cost is compared separately



Forecast, not measured interval data. Solar case uses bundled SDG&E EV-T00-5; monthly values precede credit banking/true-up. Normal weather: 1991–2020 ERA5. Solar: PVGIS NSRDB 2005–2015. No gas charges or climate credits modeled.

Load estimate from the six bills

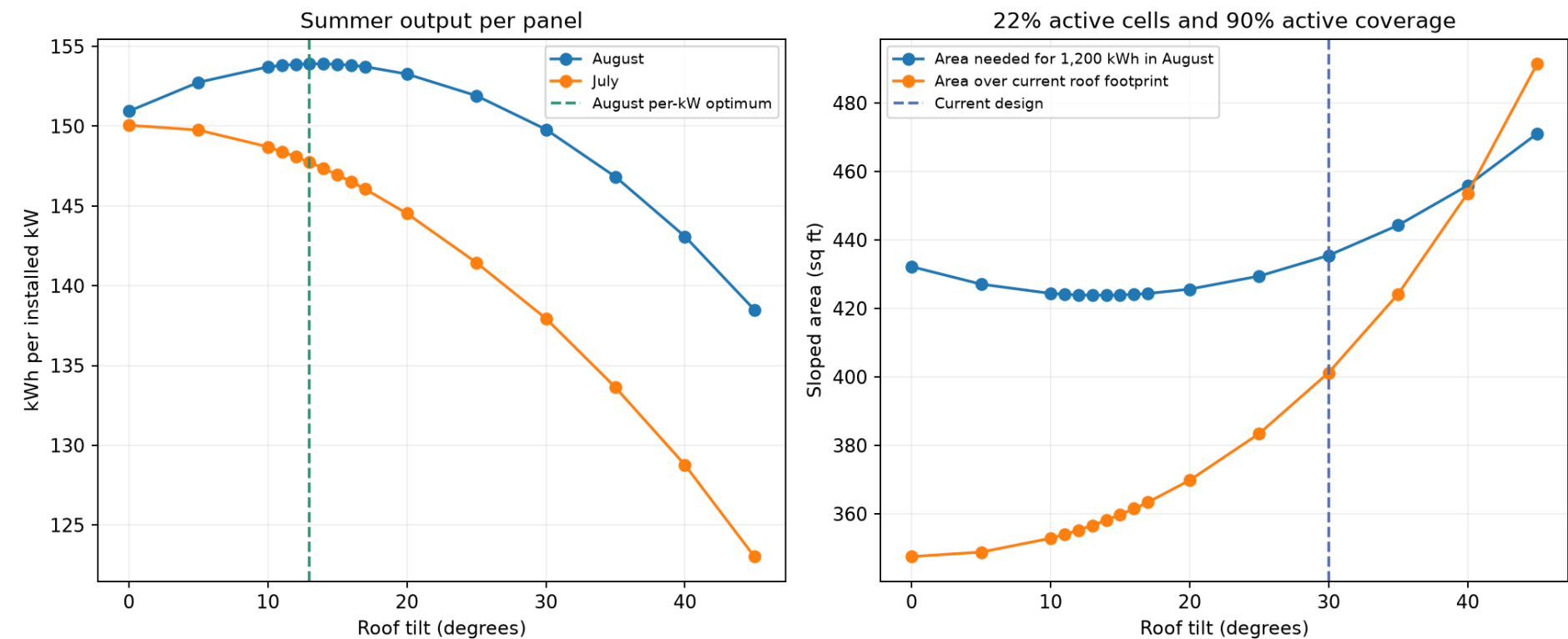
February establishes a steady load of $768/28 = 27.43$ kWh/day, or 1.14 kW continuously. The fitted cooling model is $\text{daily kWh} = 27.43 + 1.53 \times \max(\text{mean temperature} - 63.6^\circ\text{F}, 0)$. Its error over the six bills is 56 kWh/month. April is the largest mismatch; real billing dates, occupancy and non-cooling loads are unavailable.



Bill	Mean °F	Actual kWh	Model kWh	Residual
2026-02	60.4	768	768	+0
2026-03	64.7	874	904	-30
2026-04	63.0	712	823	-111
2026-06	66.2	997	943	+54
2026-07	71.4	1,179	1,223	-44
2026-08	74.9	1,409	1,388	+21

Normal-year estimate: 11,096 kWh/year. The warmer 2016–2025 average gives 11,335 kWh/year. August 2026 was 4.3°F warmer than the 1991–2020 ERA5 average, consistent with the exceptional 1,409 kWh bill.

Angle, area and summer production



Tilt	Jul / kW	Aug / kW	kW for Aug target	Roof needed at 90%	Roof over footprint	Aug max on footprint
0°	150.1	150.9	7.95	432 ft²	348 ft²	965 kWh
13°	147.7	153.9	7.80	424 ft²	357 ft²	1010 kWh
20°	144.5	153.3	7.83	426 ft²	370 ft²	1043 kWh
30°	137.9	149.8	8.01	435 ft²	401 ft²	1106 kWh
35°	133.6	146.8	8.17	444 ft²	424 ft²	1146 kWh
40°	128.8	143.1	8.39	456 ft²	454 ft²	1194 kWh
45°	123.0	138.5	8.66	471 ft²	491 ft²	1252 kWh

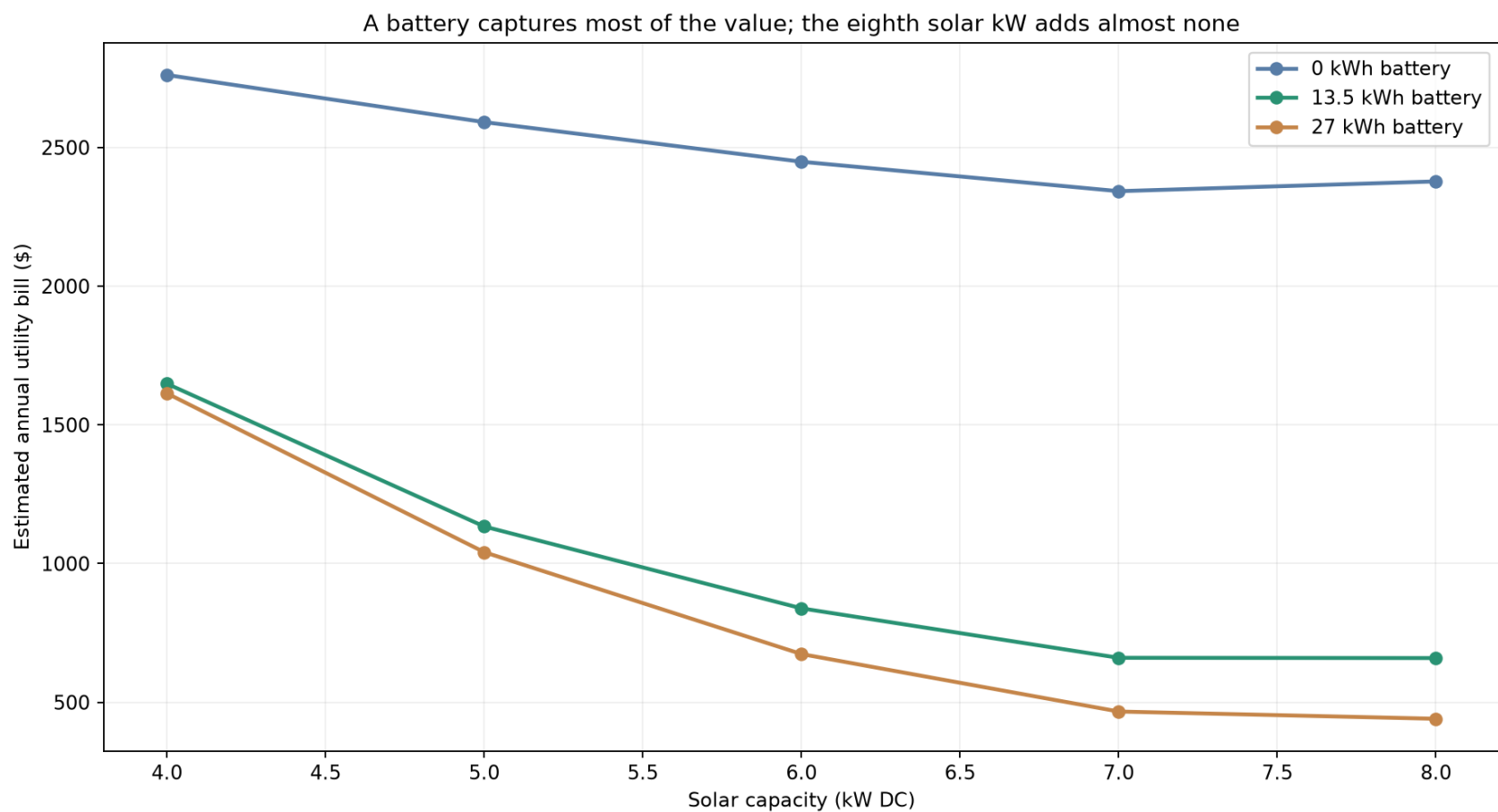
At fixed 22% cell efficiency and 90% active coverage, approximately 40° is required to approach 1,200 kWh on the present footprint. That changes the building height and roof geometry. Keeping 30° and using 23–24% cells or adding roughly 34 sq ft of solar slope is the less disruptive path. A southwest azimuth can add about 1.7% in August in this historical weather model, but it reduces annual output and cannot be achieved by a roof-integrated array unless the solar plane itself is reoriented.

Monthly energy backfill

Month	Normal mean °F	Use	7.80 kW @ 13°	8.01 kW @ 30°	7 kW @ 30°
Jan	56.1	850	763	944	825
Feb	56.6	768	826	967	845
Mar	58.2	850	1,095	1,195	1,044
Apr	60.1	823	1,169	1,193	1,043
May	62.5	850	1,190	1,157	1,011
Jun	65.2	896	1,166	1,106	967
Jul	68.9	1,105	1,152	1,105	966
Aug	70.5	1,182	1,200	1,200	1,049
Sep	70.0	1,117	1,052	1,118	977
Oct	66.3	982	939	1,070	935
Nov	61.0	823	796	977	854
Dec	56.1	850	712	903	789

The 13° summer-target system produces about 12,059 kWh/year; the 30° target system produces about 12,936 kWh/year. PVGIS includes historical cloud and irradiance variation. ERA5 1991–2020 mean cloud cover was approximately 50% in February, 31% in July, and 24% in August. On August 15, the modeled sun is above the horizon from about 6:12 a.m. to 7:34 p.m.; July 15 is about 5:51 a.m. to 7:58 p.m.

System economics



30° system	Installed assumption	Utility bill	First-year saving	Annualized equipment	Levelized net
4 kW + 0 kWh	\$20,240	\$2,761	\$1,542	\$1,556	\$-84
6 kW + 0 kWh	\$25,360	\$2,449	\$1,855	\$1,944	\$-175
5 kW + 13.5 kWh	\$37,050	\$1,134	\$3,169	\$3,053	\$-29
6 kW + 13.5 kWh	\$39,610	\$839	\$3,465	\$3,247	\$58
7 kW + 13.5 kWh	\$42,170	\$660	\$3,643	\$3,442	\$33
8 kW + 13.5 kWh	\$44,730	\$659	\$3,644	\$3,636	\$-160
7 kW + 27 kWh	\$53,920	\$467	\$3,837	\$4,516	\$-856

Why the battery matters: new Solar Billing Plan imports and exports are valued hourly, and SDG&E's 4–9 p.m. imports are expensive. The model charges the battery only from surplus solar and uses it against the most valuable imports. It does not assume grid arbitrage. Export and delivery credit buckets are modeled separately; annual net surplus is reduced to net-surplus compensation.

Sensitivity and decision

Scenario	Selected system	Annualized net	25-year NPV
Custom roof, central	6 kW + 13.5 kWh	\$58	\$815
Conventional installation	6 kW + 13.5 kWh	\$590	\$8,315
Custom premium \$15,000	6 kW + 13.5 kWh	\$-474	\$-6,685
Installed cost -20%	6.5 kW + 13.5 kWh	\$722	\$10,179
Installed cost +20%	3.5 kW + 0 kWh	\$-377	\$-5,314
Discount rate 3%	6.5 kW + 13.5 kWh	\$561	\$9,776
Discount rate 8%	3 kW + 0 kWh	\$-491	\$-5,244
Conditional SDCP rebate	6 kW + 13.5 kWh	\$393	\$5,540
Battery replacement full price	4 kW + 0 kWh	\$-84	\$-1,189
Recent 2016-25 climate	6 kW + 13.5 kWh	\$98	\$1,385
Cooling +50%	7 kW + 13.5 kWh	\$135	\$1,898
Base computers -20%	4 kW + 0 kWh	\$-224	\$-3,156
Base computers +20%	7 kW + 13.5 kWh	\$339	\$4,774
PV weather -10%	7 kW + 13.5 kWh	\$-79	\$-1,112
PV weather +10%	6 kW + 13.5 kWh	\$167	\$2,353
Battery usable capacity -20%	6 kW + 13.5 kWh	\$-64	\$-907
Import rates -20%	4 kW + 0 kWh	\$-364	\$-5,136
Import rates +20%	6 kW + 13.5 kWh	\$710	\$10,004
Export credits halved	6 kW + 13.5 kWh	\$35	\$498
AC delayed four hours	6 kW + 13.5 kWh	\$77	\$1,079

The recommendation is robust in shape, but not in profitability: 5–7 kW plus one battery is usually the useful range, while a second battery is too expensive and the eighth solar kilowatt adds little bill value. Profitability changes sign with installation cost, real discount rate, utility rates, and battery aging. Obtain quotes that separately price (1) the waterproof roof, fascia and architectural solar skin, (2) active PV hardware, and (3) battery and controls. Treating all custom-roof work as a solar cost makes the project approximately break-even; treating that finish as required building work makes solar economically attractive.

A conditional San Diego Community Power battery rebate improves the modeled 6 kW + one-battery case, but it is not included in the central result because the bill's generation provider and rate plan have not been confirmed. The current federal homeowner Residential Clean Energy Credit is unavailable for systems placed in service after December 31, 2025, so no 30% federal credit is deducted.

Model boundaries and data sources

This is a planning model, not a contractor proposal or utility bill guarantee. The load model has six monthly observations and no interval data. Calendar months substitute for billing periods. Air-conditioning timing is synthesized from hourly temperature. Gas, climate credits, taxes, minimum charges and unknown CCA generation rates are excluded. Results currently use bundled SDG&E; EV-TOU-5 rates effective August 1, 2026 and the NBT00 records actually present in SDG&E;'s downloaded 'Current 2026' export file. SDG&E;'s readme describes NBT26, creating a source inconsistency that should be resolved against the eventual interconnection application.

Weather: Open-Meteo ERA5 reanalysis, 1991–2026, 0.25° grid. Solar: European Commission PVGIS 5.2 with NSRDB 2005–2015, crystalline-silicon building mounting, 14% system loss, and no external horizon obstruction. Hourly dispatch uses PVGIS typical meteorological year shapes normalized to PVGIS monthly energy. Cost: September 2026 EnergySage San Diego average \$2.56/W; central battery \$14,250 for 13.5 kWh; custom solar-roof allowance \$7,500; electrical allowance \$2,500; year-15 inverter plus 60% battery replacement; 0.5%/year PV value degradation; 5% real discount rate; 25-year horizon. Utility rates and equipment prices are held constant after inflation.

Files supplied with this report: monthly.csv, bill-fit.csv, tilt-and-area.csv, azimuth.csv, system-comparison.csv, sensitivity.csv, sun-geometry.csv, and a compressed 8,760-hour audit file. The analysis program and downloaded source files are retained alongside the results.